Continuous Sign Language Pose Generation via Text-driven Diffusion Model

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***Abstract*—Sign language generation aims to generate semantically aligned sign pose sequences from text. Diffusion models have shown great potential in generation tasks and provided new ideas for high-quality sequence generation. To this end, we propose a Text-driven Diffusion Model (TDM) for sign language generation. During the training phase, TDM encodes text and incorporates it as conditional input into the diffusion model to generate sign pose sequences. To ensure the high quality and accuracy of the generated pose sequences, we use two loss functions. The joint loss $\mathcal{L}_{joint}$ is used to precisely measure and minimize the differences between the joint positions of the generated pose sequences and those of the ground truth. Similarly, the bone orientation loss $\mathcal{L}_{bone}$ is instrumental in ensuring that the orientation of the bones in the generated poses aligns with the actual and correct orientations. In the inference stage, TDM starts from a noise sequence under the strict constraints of text conditions and gradually refines and generates a semantically consistent sign pose sequence. Extensive experiments on the PHOENIX14T datasets validate the effectiveness of our method.**

Keywords-Computer vision, Natural language processing

I. INTRODUCTION

Sign language, an essential tool for seamless interaction between key research fields and the deaf community, is at the forefront of a technological revolution that significantly impacts the construction of information accessibility. Sign language is increasingly relevant to large language models [1], [2] and cross-media reasoning [3]–[9]. In the digital age, with information flowing freely in most of society, ensuring that the deaf can equally access this wealth of knowledge is not only a matter of social justice but also a frontier in academic and technological exploration.

Currently, academic research in sign language technology mainly focuses on three major interrelated directions: Sign Language Translation (SLT), Sign Language Recognition (SLR), and Sign Language Generation (SLG). SLT aims to bridge the communication gap between sign language and spoken language, promoting effective communication among the deaf and hard of hearing, while SLR focuses on the recognition of signs. Although significant progress has been made in SLT [10]–[14] and SLR [15]–[17], SLG still poses a complex and urgent challenge. The complexity of SLG lies in the unique grammatical rules of sign language. Sign language is not just a visual coun-terpart to spoken language. It is an independent language system with its own grammar, syntax, and semantics.

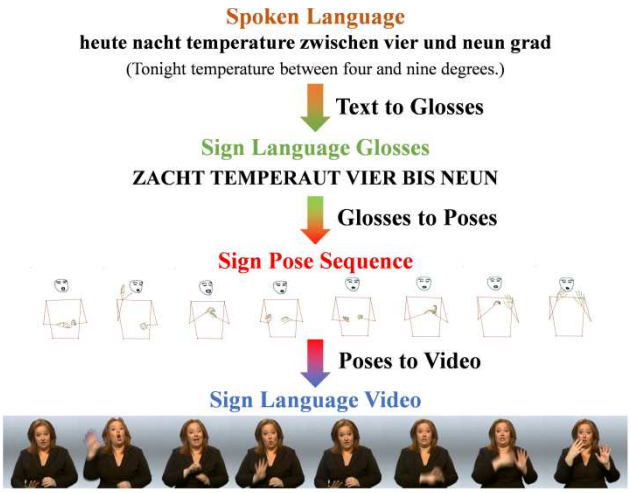


Figure 1. The task of generating sign pose sequences directly from text extremely daunting challenges. Classic pipelines for Sign Language Generation: text-to-gloss, gloss-to-pose and pose-to-video. In this work, we go beyond the commonly used pipeline and focus on the T2P task of generating sign pose sequences directly from text.

To address this challenge, existing methods typically adopt a two-step process. First, the text is converted into annotations, a representation that captures the specific meanings of gestures. Then, a sequence of signs is generated based on these annotations. However, this approach has several drawbacks. The conversion from text to annotations and then to signs adds multiple levels of complexity. Therefore,

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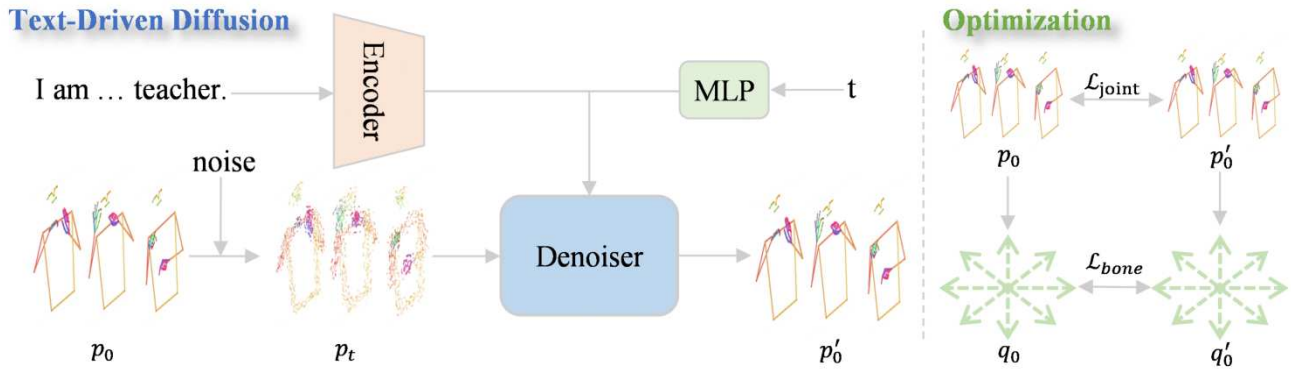


Figure 2. Overall framework of our method - TDM. It consists of a text encoder, a denoiser, and a MLP. For the optimization, we employed two main loss functions. The $\mathcal{L}_{joint}$ loss function is primarily used to constrain the joint coordinates, while the $\mathcal{L}_{bone}$ loss function is mainly applied to constrain the direction of bone movement.

we attempt to generate sign pose sequences directly from text. Our main contributions are:

- We propose an excellent baseline based on a diffusion model for the T2P task.
- Experiments on the PHOENIX14T dataset demonstrate the superiority of our method.

II. Related Work

A. Sign Language Generation

In the generation task, Variational Autoencoders (VAEs) [18] and Generative Adversarial Networks (GANs) [19] showed remarkable performance. Inspired by their performance, a series of methods based on Generative Adversarial Networks (GANs) [20], [21] and Variational Autoencoders (VAEs) [22], [23] have emerged. Although these methods attempt to control the mapping from text to visual representation by encoding textual information, they often fail to handle complex semantics effectively. The Transformer [24]–[30] represents a significant advancement. By utilizing an attention mechanism, it can simultaneously capture rich semantic information in the text from various representational subspaces. This allows it to generate more coherent and natural sign pose sequences. Nevertheless, sign language presents unique challenges because of its rich vocabulary, complex grammatical structure, and diverse expressions. To overcome these challenges, the diffusion model has been introduced. G2P-DDM [31] and Sign-IDD [27] are diffusion-based soluteions. These models generate coordinate representations of hand gestures starting from Gaussian noise. By gradually denoising the noise, they can precisely depict the spatial positions and movement trajectories of sign language gestures, holding great potential for a breakthrough in SLG.

B. Diffusion Models

The diffusion model [32] is a sophisticated generative model. It first gradually adds noise to the original data and then, through a reverse denoising process, generates high-quality samples. This unique mechanism has allowed diffusion models to make significant progress in numerous domains. For example, in the field of image generation, diffusion models have achieved remarkable results. For example, DDPM [33] and DDIM [34] have demonstrated that diffusion models are capable of generating highly realistic and diverse images. These models can capture the complex details and semantic information in natural images, thus revolutionizing the field of image generation. Diffusion models have also demonstrated great potential in text generation [35], [36]. In the area of image inpainting, diffusion models [37] have also had an impact. When given an image with missing parts, diffusion-based inpainting algorithms can accurately predict and fill in the missing regions, thereby restoring the image to its original or a plausible state. Moreover, in the field of video generation, diffusion models [38] and [39] have been able to generate smooth and visually appealing videos while taking into account the continuity of frames. However, despite their extensive success in these areas, the applications of diffusion models in sign language are still relatively limited. Therefore, in this study, we aim to bridge this gap by exploring the potential of propagation models in the SLG.

III. Methodology

As shown in Figure 2, the TDM mainly comprises three components: 1) the forward process of gradually introducing noise into the sign pose sequences; 2) the reverse process that recovers the original sign pose sequences from the noisy pose sequences through denoising; and 3) the denoiser $\mathcal{D}$.

A. Forward Process

The forward process begins with the original sign pose sequence p_0. In this process, noise is added to p_0 successively in t steps, where $t \in T$. Eventually, this results in the generation of the noisy sign pose sequence p_t. The formula is as follows:

$$p_t = \gamma_t + \sigma_t \epsilon, \tag{1}$$

where ϵ is sampled from a Gaussian distribution. The γ_t and σ_t represent the ratio of information p_0 and noise ϵ, respectively, and $\gamma_t^2 + \sigma_t^2 = 1$ [33], [34].

B. Reverse Process

During training, we use a denoiser $\mathcal{D}$, to directly predict the original sign pose sequences, rather than the noise [33], [34]. The relevant condition g plays a vital role in guiding the denoising process. The denoising operation is expressed by the following formula:

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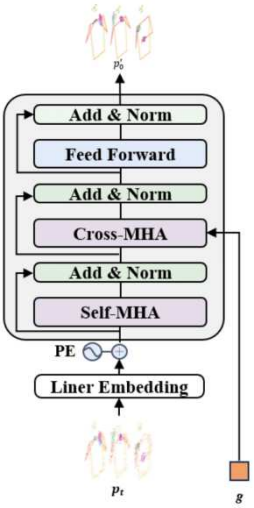


Figure 3. Detailed implementation details of our denoiser $\mathcal{D}$.

$$p_0' = \mathcal{D}(p_t, g), \tag{2}$$

where p_t represents the data that is contaminated by noise. The relevant condition g is generated by the text and the diffusion time step t.

During the inference phase, we initialize the sign pose sequence p_T by sampling noise from a standard Gaussian distribution. This initialized pose sequence p_T and the associateed condition g are fed into the denoiser $\mathcal{D}$ to predict the original sign pose sequence $\tilde{p}_{t'}$ at the next time step, which serves as the input of DDIM [34]. The formula is as follows:

$$\tilde{p}_{t'} = \gamma_{t'}\tilde{p}_0 + \sigma_{t'}\epsilon, \tag{3}$$

where t' represents the next step after T. $\epsilon \sim \mathcal{N}(0,1)$ is standard Gaussian noise independent of the input p_T. Finally, we use $\tilde{p}_{t'}$ as the input to the denoiser $\mathcal{D}$ and re-predict the output $\tilde{p}_0$ with the relevant condition g. This process will be iterated i times.

C. Denoiser

To comprehensively and clearly understand the denoiser $\mathcal{D}$, a systematic exploration beginning with its training phase is essential.

The first is the Linear Embedding Layer (LE), which transforms the input sequence into a representation that is more suitable for subsequent processing. The formula for the Linear Embedding Layer is as follows:

$$p_u = W^p \cdot p_t + b^p, \tag{4}$$

where W^p represents the weight matrix, and b^p is the bias term.

After the Linear Embedding, we use positional encoding (PE) to enhance the temporal characteristics of the transformed sequence. The positional encoding process is defined by the following formula:

$$\hat{p}_u = p_u + PE(n), \tag{5}$$

where the PE function is a predefined sinusoidal function.

With the linearly embedded and position-encoded sequence $\hat{p}_u$ and the relevant condition g in hand, we move on to the final processing stage. This stage utilizes two powerful attention mechanisms: multi-head attention and cross-attention mechanisms. The core of the multi-head attention mechanism is based on the following set of equations:

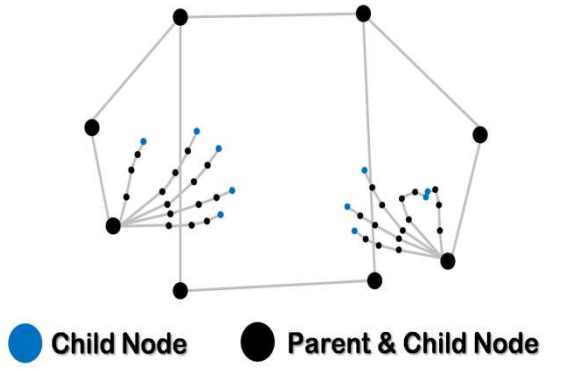


Figure 4. Parent-child node partition relationship on the PHOENIX14T dataset.

$$Attention(Q,K,V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V, \tag{6}$$

$$head_i = Attention(QW_i^Q, KW_i^K, VW_i^V), \tag{7}$$

$$MHA(Q,K,V) = (head_1, \ldots, head_h) \cdot W. \tag{8}$$

D. Loss function

In our framework, we employ two distinct loss functions to boost the model's performance. Specifically, the $\mathcal{L}_{joint}$ is utilized to enforce the accuracy of the joints, while the $\mathcal{L}_{bone}$ serves to constrain the orientation of the bones.

Joint Constraint: To guaranty an exact alignment with the ground truth, we use the joint loss function to enforce the accuracy of joint positions in the pose. The joint constraint $\mathcal{L}_{joint}$ is defined as follows:

$$\mathcal{L}_{joint} = \frac{1}{J}\sum_{j=1}^{J}|p_j - p_j'|, \tag{9}$$

where J represents the total number of joints. This formula measures the error in joint positions by computing the absolute difference between the predicted joint position $p_{j'}$ and the true joint position p_j, and then taking the average across all joints.

Bone Constraint: To enhance the accuracy of bone orientation in the generated poses, we introduce $\mathcal{L}_{bone}$. The formula is as follows:

$$\mathcal{L}_{bone} = \frac{1}{B}\sum_{b=1}^{B}|q_b - q_b'|, \tag{10}$$

where q_b and q_b' represent the bone orientations obtained from the ground truth and the predicted results p_0', and B represents the number of bones. As shown in Figure 4, we calculate the direction of bone movement by dividing the parent and child nodes. The specific formula is as follows:

$$q = (\vec{x}_b, \vec{y}_b, \vec{z}_b). \tag{11}$$

By computing the sum of squared differences between the actual and predicted bone directions and then averaging these, we can effectively quantify the error in bone

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Figure 5. Visualization examples of generated pose sequences on PHOENIX14T. We compare TDM with ground-truth.

orientation, thereby enhancing the accuracy of bone orientation.

The final optimization goal is:

$$\mathcal{L} = \mathcal{L}_{joint} + \lambda \mathcal{L}_{bone}, \tag{12}$$

where λ is 0.1. Please note that the $\mathcal{L}_{bone}$ is only applicable to the body and not to the facial area.

IV. Experiments

A. Datasets

Our experiments primarily used the PHOENIX14T dataset [42] from the CVPR 2025 SLRTP challenge [41]. This comprehensive dataset contains 8,257 complete facial and gesture instances, covering 2,887 distinct German words and 1,066 gesture types. All test results are based on the hidden test set of PHOENIX14T.

B. Implementation Details

Regarding the model architecture, the text encoder within our framework has the same structure as the Progressive Transformer [40].

Concerning the model parameters, in our proposed method, both the text encoder and the denoiser $\mathcal{D}$ are set up with 4 layers and 8 heads, sharing an embedding dimension of 1024. These parameter settings were meticulously chosen through preliminary experiments and analyzes. The 4-layer architecture allows for an adequate depth of feature extraction, whereas the 8-head mechanism empowers the model to concurrently capture different aspects of the input data. An embedding dimension of 1024 offers a high-dimensional representation of the data, facilitating better encoding and subsequent processing.

TABLE I. The summary of hidden test set results

Methods	B1↑	B4↑	ROUGE↑	WER↓	DTW-MJE
PT	18.33	0.78	21.10	141.93	**0.0467**
Hacettepe	27.51	2.91	27.29	**105.49**	0.0531
Ours	**32.54**	**5.66**	**32.92**	107.93	0.0492

Our diffusion model is equipped with a cosine scheduler. The scheduler sets $T = 1000$. The noise injection operations are crucial for the diffusion model to learn the data distribution in a noisy setting. Moreover, in the inference stage $i = 5$. This number of inferences has been determined to strike a balance between computational cost and the accuracy of the generated outcomes.

C. Comparison with other methods

The experimental results on the hidden test set are summarized in Table I. We conducted a comprehensive

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comparison with other state-of-the-art methods, such as PT [40] and Hacettepe [41], using evaluation metrics including BLEU-1, BLEU-4, ROUGEF, Word Error Rate (WER), and DTW-MJE. The results demonstrate that our proposed method achieves the best performance on multiple metrics. Specifically, it scores 30.54, 5.66, and 32.92 on BLEU-1, BLEU-4, and ROUG, respectively, significantly outperforming the other compared methods. Although it is slightly lower than some methods on WER and DTW-MJE, the overall performance indicates that our model possesses a clear advantage in terms of generation quality and semantic consistency.

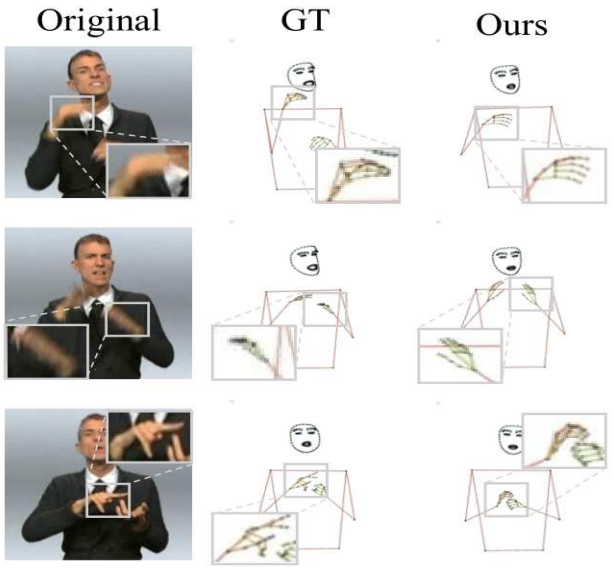


Figure 6. Visualization examples of generated pose sequences on PHOENIX14T. We compare TDM with ground-truth.

D. Qualitative Results

Visualization of sign pose sequences. Figure 5 presents qualitative comparisons of generated pose sequences on the PHOENIX14T dataset. Each example consists of an input German sentence (Text), the corresponding original sign video frames (Original), the ground-truth pose sequences extracted from the original signer (GT), and the poses generated by our model (Ours). From the visualization, we observe that our generated poses exhibit high temporal consistency and semantic alignment with the input text. Compared to the ground-truth poses, our results preserve correct global body structure while accurately modeling key motion details such as hand trajectories and arm orientations. Moreover, the generated pose sequences follow the natural rhythm of signing, demonstrating smooth temporal transitions without unrealistic jitter or collapse. These results highlight the capability of our method to produce interpretable and realistic sign pose sequences conditioned on text.

Visualization of fine-grained Pose. Figure 6 provides a detailed qualitative comparison between the ground-truth poses (GT) and the poses generated by our model (Ours) on PHOENIX14T. To better illustrate key motion accuracy, we zoom in on the hand regions, which are crucial for conveying semantic information in sign language. Compared with GT, our method generates plausible and well-structured hand configurations, maintaining correct spatial relations between joints. The generated poses accurately preserve fine-grained motion directions and signing dynamics, especially in hand orientation and wrist articulation. Furthermore, our model avoids unnatural joint distortions and produces smoother limb trajectories than conventional baselines. These visual results demonstrate that our approach can effectively capture both global body structure and local gesture details essential for sign language understanding.

V. CONCLUSIONS

We propose a sign language generation method based on a text-driven diffusion model (TDM). During the training phase, this method encodes text as conditional input and generates sign language pose sequences using a diffusion model. Extensive experimental results on the PHOENIX14T dataset demonstrate that the proposed method can effectively generate high-quality, semantically consistent sign language sequences.

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